

Commerce XI Maths App

Theory (Pages 27–30)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 40

Question 1 - Theory P27

Range and quartile deviation are based mainly on

Options

- A. two values
- B. all observations
- C. only the mean
- D. only class width

Correct answer: two values

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives two values.

Question 2 - Theory P27

Variance and standard deviation use

Options

- A. all observations
- B. only the largest value
- C. only quartiles
- D. only class limits

Correct answer: all observations

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives all observations.

Question 3 - Theory P27

Variance is denoted by

Options

- A. $\text{Var}(X)$ or σ^2
- B. σ only
- C. $\bar{x}$ only
- D. Q.D.

Correct answer: $\text{Var}(X)$ or σ^2

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\text{Var}(X)$ or σ^2 .

Question 4 - Theory P27

Variance is the arithmetic mean of

Options

- A. squares of deviations from the arithmetic mean
- B. absolute observations
- C. class widths
- D. quartiles

Correct answer: squares of deviations from the arithmetic mean

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives squares of deviations from the arithmetic mean.

Question 5 - Theory P27

Standard deviation is denoted by

Options

- A. σ
- B. σ^2
- C. $\bar{x}$
- D. N

Correct answer: σ

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives σ .

Question 6 - Theory P27

Standard deviation is the

Options

- A. positive square root of variance
- B. square of variance
- C. negative square root of variance
- D. same as the mean

Correct answer: positive square root of variance

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives positive square root of variance.

Question 7 - Theory P27

Variance can never be

Options

A. negative

B. zero

C. positive

D. a decimal

Correct answer: negative

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives negative.

Question 8 - Theory P27

If every observation is identical, variance equals

Options

A. 0

B. 1

C. the common value

D. N

Correct answer: 0

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 0.

Question 9 - Let's Derive P27

For raw data, variance can be calculated as

Options

A. $(\sum x_i^2)/(n) - \bar{x}^2$

B. $(\sum x_i)/(n) + \bar{x}^2$

C. $\sum x_i^2 - \bar{x}$

D. $\bar{x}^2$ only

Correct answer: $(\sum x_i^2)/(n) - \bar{x}^2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(\sum x_i^2)/(n) - \bar{x}^2$.

Question 10 - Theory P27

For an ungrouped frequency distribution, total frequency is

Options

- A. $N = \sum f_i$
- B. $N = \sum x_i$
- C. $N = \bar{x}$
- D. $N = \sigma^2$

Correct answer: $N = \sum f_i$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $N = \sum f_i$.

Question 11 - Theory P27

For frequency data, the mean is

Options

- A. $(\sum f_i x_i) / (N)$
- B. $(\sum x_i) / (N)$
- C. $(\sum f_i) / (N)$
- D. $\sum f_i x_i$

Correct answer: $(\sum f_i x_i) / (N)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(\sum f_i x_i) / (N)$.

Question 12 - Theory P27

For frequency data, variance by the direct method is

Options

- A. $(\sum f_i x_i^2) / (N) - \bar{x}^2$
- B. $(\sum f_i x_i) / (N) - \bar{x}$
- C. $(\sum x_i^2) / (N)$
- D. $(\sum f_i) / (N)$

Correct answer: $(\sum f_i x_i^2) / (N) - \bar{x}^2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(\sum f_i x_i^2) / (N) - \bar{x}^2$.

Question 13 - Theory P27-28

For grouped data, x_i represents the

Options

- A. class midpoint
- B. class frequency
- C. lower boundary
- D. cumulative frequency

Correct answer: class midpoint

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives class midpoint.

Question 14 - Theory P28

Variance is independent of a change of

Options

- A. origin
- B. scale
- C. unit
- D. class width

Correct answer: origin

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives origin.

Question 15 - Theory P28

Variance is not independent of a change of

Options

- A. scale
- B. origin
- C. sample size
- D. frequency total

Correct answer: scale

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives scale.

Question 16 - Theory P28

If $d = X - A$, then

Options

- A. $\sigma_x^2 = \sigma d^2$
- B. $\sigma_x^2 = A\sigma d^2$
- C. $\sigma_x^2 = \sigma d$
- D. $\sigma_x = A$

Correct answer: $\sigma_x^2 = \sigma d^2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\sigma_x^2 = \sigma d^2$.

Question 17 - Theory P28

If $u = (X - A)/(h)$, then

Options

- A. $\sigma_x^2 = h^2\sigma_u^2$
- B. $\sigma_x^2 = h\sigma_u^2$
- C. $\sigma_x^2 = (\sigma_u^2)/(h)$
- D. $\sigma_x^2 = A^2\sigma_u^2$

Correct answer: $\sigma_x^2 = h^2\sigma_u^2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\sigma_x^2 = h^2\sigma_u^2$.

Question 18 - Theory P28

If $u = (X - A)/(h)$, standard deviations satisfy

Options

- A. $\sigma_x = h\sigma_u$
- B. $\sigma_x = h^2\sigma_u$
- C. $\sigma_x = (\sigma_u)/(h^2)$
- D. $\sigma_x = A\sigma_u$

Correct answer: $\sigma_x = h\sigma_u$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\sigma_x = h\sigma_u$.

Question 19 - Theory P29

The shortcut method is also called the

Options

- A. step-deviation method
- B. quartile method
- C. range method
- D. graphical method

Correct answer: step-deviation method

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives step-deviation method.

Question 20 - Theory P29

In $u = (X - A)/(h)$, A is the

Options

- A. assumed mean
- B. actual variance
- C. total frequency
- D. standard deviation

Correct answer: assumed mean

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives assumed mean.

Question 21 - Theory P29

In $u = (X - A)/(h)$, h is generally a common

Options

- A. class width or scale factor
- B. frequency
- C. mean
- D. quartile

Correct answer: class width or scale factor

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives class width or scale factor.

Question 22 - Solved Example P28

For 9, 12, 15, 18, 21, 24, 27, the mean is

Options

- A. 18
- B. 15
- C. 21
- D. 36

Correct answer: 18

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 18.

Question 23 - Solved Example P28

For 9, 12, 15, 18, 21, 24, 27, the variance is

Options

- A. 36
- B. 6
- C. 18
- D. 252

Correct answer: 36

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 36.

Question 24 - Solved Example P28

For 9, 12, 15, 18, 21, 24, 27, the standard deviation is

Options

- A. 6
- B. 36
- C. 3
- D. 18

Correct answer: 6

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 6.

Question 25 - Solved Example P28-29

For marks 10, 13, 17, 20, 23, the mean is

Options

- A. 16.6
- B. 16
- C. 17
- D. 21.84

Correct answer: 16.6

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 16.6.

Question 26 - Solved Example P28-29

For marks 10, 13, 17, 20, 23, the variance is

Options

- A. 21.84
- B. 16.6
- C. 4.67
- D. 109.2

Correct answer: 21.84

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 21.84.

Question 27 - Solved Example P28-29

For marks 10, 13, 17, 20, 23, the standard deviation is approximately

Options

- A. 4.67
- B. 21.84
- C. 16.6
- D. 5.84

Correct answer: 4.67

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 4.67.

Question 28 - Solved Example P29

A die is rolled 30 times with frequencies 2, 6, 2, 5, 10, 5 for scores 1–6. The mean score is

Options

- A. 4
- B. 3
- C. 4.5
- D. 5

Correct answer: 4

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 4.

Question 29 - Solved Example P29

A die is rolled 30 times with frequencies 2, 6, 2, 5, 10, 5 for scores 1–6. The variance is approximately

Options

- A. 2.47
- B. 1.57
- C. 4
- D. 18.47

Correct answer: 2.47

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 2.47.

Question 30 - Solved Example P29

A die is rolled 30 times with frequencies 2, 6, 2, 5, 10, 5 for scores 1–6. The standard deviation is approximately

Options

- A. 1.57
- B. 2.47
- C. 4
- D. 0.57

Correct answer: 1.57

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 1.57.

Question 31 - Solved Example P29

For $x = 15, 20, 25, 30, 35, 40, 45$ and frequencies $13, 12, 15, 18, 17, 10, 15$, the coded mean $\bar{u}$ is

Options

- A. 0.04
- B. 0.4
- C. 4
- D. -0.04

Correct answer: 0.04

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 0.04.

Question 32 - Solved Example P29

For $x = 15, 20, 25, 30, 35, 40, 45$ and frequencies $13, 12, 15, 18, 17, 10, 15$, $\text{Var}(u)$ is approximately

Options

- A. 3.7184
- B. 0.04
- C. 2.7824
- D. 18.592

Correct answer: 3.7184

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 3.7184.

Question 33 - Solved Example P29

For $x = 15, 20, 25, 30, 35, 40, 45$ and frequencies $13, 12, 15, 18, 17, 10, 15$, with $h = 5$, the variance of X is

Options

- A. 92.96
- B. 18.592
- C. 3.7184
- D. 25

Correct answer: 92.96

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 92.96.

Question 34 - Solved Example P29

For $x = 15, 20, 25, 30, 35, 40, 45$ and frequencies $13, 12, 15, 18, 17, 10, 15$, with $h = 5$, the standard deviation of X is approximately

Options

- A. 9.64
- B. 92.96
- C. 18.59
- D. 5

Correct answer: 9.64

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 9.64.

Question 35 - Solved Example P30

Class	45-55	55-65	65-75	75-85	85-95	95-105	105-115	115-125
Frequency	7	20	27	23	13	6	3	1

For this grouped distribution, $\text{Var}(u)$ is

Options

- A. 2.25
- B. 1.5
- C. 4.5
- D. 225

Correct answer: 2.25

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 2.25.

Question 36 - Solved Example P30

Class	45-55	55-65	65-75	75-85	85-95	95-105	105-115	115-125
Frequency	7	20	27	23	13	6	3	1

[[line:With $A = 90$ and $h = 10$, $\text{Var}(X)$ is]]

Options

- A. 225
- B. 22.5
- C. 15
- D. 2.25

Correct answer: 225

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 225.

Question 37 - Solved Example P30

Class	45-55	55-65	65-75	75-85	85-95	95-105	105-115	115-125
Frequency	7	20	27	23	13	6	3	1

The standard deviation is

Options

- A. 15
- B. 225
- C. 10
- D. 2.25

Correct answer: 15

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 15.

Question 38 - Solved Example P30

Height of plants (cm)	20-25	25-30	30-35	35-40	40-45	45-50
No. of plants	145	125	90	40	45	55

[[line:With $A = 32.5$ and $h = 5$, $\text{Var}(u)$ is approximately]]

Options

- A. 2.7824
- B. 0.24
- C. 2.25
- D. 69.56

Correct answer: 2.7824

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 2.7824.

Question 39 - Solved Example P30

Height of plants (cm)	20-25	25-30	30-35	35-40	40-45	45-50
No. of plants	145	125	90	40	45	55

The variance of X is approximately

Options

A. 69.56

B. 8.34

C. 2.7824

D. 13.90

Correct answer: 69.56

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 69.56.

Question 40 - Solved Example P30

Height of plants (cm)	20-25	25-30	30-35	35-40	40-45	45-50
No. of plants	145	125	90	40	45	55

The standard deviation is approximately

Options

A. 8.34

B. 69.56

C. 5

D. 2.78

Correct answer: 8.34

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 8.34.